

Addressing GNSS Vulnerabilities in AAM: A Multi-Modal UAV Testbed for Redundant and Reliable Navigation

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Advanced Air Mobility (AAM) - the idea of extending the current mobility landscape by transporting people and goods flexibly from A to B through the sky at low altitudes receives increasing interest in industry and academia. A core element are electrical-driven, highly automated aircrafts which are capable of vertical take-off and landing (commonly referred to as eVTOLs) (Xiang et al., 2024). One of the key challenge for the safe operation of eVTOLs, generally small-scale UAVs, is the precise and reliable positioning and navigation.

UAVs which are used for various tasks such as mapping, precision agriculture, and search and rescue commonly rely on the complementary fusion of GNSS and inertial sensors (IMU). The IMU measurements are integrated to provide the system's velocity, position, and attitude with a high temporal resolution. However, overtime factors such as integration errors, sensor bias, and sensor noise lead to a drift in pose estimates. This drift can be corrected by the less frequent but accurate GNSS measurements. Nevertheless, the dependence on external satellite signals makes GNSS positioning vulnerable to intentional interference, namely jamming and spoofing. Furthermore, the performance can significantly degrade through signal shadowing, multi-path interference and non-line-of-sight (Couturier and Akhloufi, 2024). Therefore, alternative navigation approaches are required for safe operations.

A well-established approach is Visual Inertial Odometry (VIO) which uses one or more cameras and IMU measurements for UAV pose estimation. While accurate in ideal conditions the use of RGB cameras has known limitations under insufficient availability of features and illumination. To tackle these drawbacks, (Vidal et al., 2018) propose a hybrid VIO pipeline that additionally fuses measurements from event cameras to cope well with such scenarios. A similar approach is LiDAR Inertial Odometry (LIO), leveraging a LiDAR sensor to provide depth information in an illumination-independent way (Wu et al., 2024). All these dead-reckoning techniques tend to accumulate drift. A common mitigation strategy, SLAM - odometry plus loop closure - relies on revisiting and recognizing places, undesirable in AAM scenarios. An alternative place recognition in the form of absolute visual localization - given a georeferenced map - is capable of drift correction even in GNSS-denied conditions.

Individual sensors and their associated approaches have their shortcomings, therefore we seek a multi-modal approach to fuse data from GNSS, IMU, cameras, and LiDAR. Redundancy with multiple sensors circumvents single point of failure, enabling

safe operations. While well-engineered commercial-off-the-shelf drones are increasingly available, they are not designed to integrate this multitude of sensors, nor are they designed to process large volumes of data in real-time directly on board. However, designing UAVs entirely from scratch is a complex and time-intensive endeavour. As a practical compromise, we augment commercially available development kits with additional peripherals, enabling both flexibility in sensor integration and sufficient onboard computational capabilities.

Our hardware design choices are driven by the SWaP-C (size, weight, power, and cost) principle to ensure suitability for UAV operations. Key specifications such as sensor range, resolution, and frame rate are selected to meet the operational requirements of our intended flight scenarios, including the maximum legal flight height of 120 meters in the EASA open category (European Union Aviation Safety Agency, 2024). To support high-bandwidth data acquisition, time synchronization, and power - ethernet with PTP and PoE accommodates over a single interface. We predominantly aim to leverage open-source solutions in our hard- and software setup. In particular, we opt for a symbiotic setup with the Pixhawk flight controller, PX4 Autopilot, QGroundControl ground control station, and other standards maintained by the Dronecode Foundation (Dronecode Foundation, 2021) as well as ROS (Macenski et al., 2022) for our testbed.

In this paper, we present a detailed overview of our hardware and software design choices. Furthermore, we focus on the sensor integration on the UAV and the calibration processes necessary for accurate data fusion - including both extrinsic calibration between sensors and intrinsic calibration of the RGB and event cameras. Finally, a unified map representation based on multi-modal sensor data, which is memory-efficient, real-time updatable, and includes probabilistic elements for sensor fusion, is desirable for high-level task planning. This common map representation can be transmitted via a wireless communication link from the UAV to the ground control station, and allows efficient real-time visualization during operations.

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